PLANT PATHOLOGY

Targeted MYC2 stabilization confers citrus Huanglongbing resistance

Pingzhi Zhao[1,2]*†, Huan Yang[1,2]†, Yanwei Sun[1]†, Jingyin Zhang[1], Kaixing Gao[1,2], Jinbao Wu[1], Chengrong Zhu[1], Cece Yin[1], Xiaoyue Chen[1], Qi Liu[1], Qiudong Xia[1,2], Qiong Li[1], Han Xiao[2,3], Hai-Xi Sun[3], Xiaoxiao Zhang[4], Long Yi[5], Changyong Zhou[6], Daniel J. Kliebenstein[7], Rongxiang Fang[1,2], Xuefeng Wang[6]*, Jian Ye[1,2]*

Huanglongbing (HLB) is a devastating citrus disease. In this work, we report an HLB resistance regulatory circuit in *Citrus* composed of an E3 ubiquitin ligase, PUB21, and its substrate, the MYC2 transcription factor, which regulates jasmonate-mediated defense responses. A helitron insertion in the *PUB21* promoter introduced multiple MYC2-binding cis-elements to create a regulatory circuit linking the PUB21 activity with MYC2 degradation. Ectopic expression of a natural dominant-negative *PUB21* paralog discovered in distant *Citrus* relatives stabilized MYC2 and conferred resistance to HLB. Antiproteolysis peptides (APPs), identified by artificial intelligence, stabilized MYC2 by binding and inhibiting PUB21 activity. A 14–amino acid peptide, APP3-14, molecularly controlled HLB in greenhouse and field trials. This approach represents a strategy to combat uncultivable pathogens through targeted disease resistance protein stabilization.

Huanglongbing (HLB), also known as citrus greening, causes the most serious economic losses in the global citrus industry. HLB is primarily caused by the phloem-colonizing bacterium *Candidatus Liberibacter asiaticus* (*C*Las) and is mainly transmitted by the Asian citrus psyllid (*Diaphorina citri*) (*1*). Effective treatments or resistance (*R*) genes have not been identified, and all commercial citrus varieties are susceptible to HLB (*1*–*3*). An emerging alternative strategy to provide resistance involves disrupting disease susceptibility (*S*) genes (*4*, *5*). In this study, we identified HLB *S* genes and developed approaches to inactivate these *S* genes for citrus resistance breeding.

PUB21 is an *S* gene for citrus HLB

The HLB pathogen *C*Las infects all *Citrus* plants, with a wide host range across the Aurantioideae subfamily of the Rutaceae family. Within this range, some Rutaceae lineages, such as Clauseneae and Rutoideae, display tolerance or resistance to HLB (Fig. 1A) (*6*). The HLB-tolerant species—the wild citrus species Mangshanyegan (*Citrus mangshanensis*) and Daoxianyeju (*Citrus daoxianensis*)—have differential regulation of the 26S proteasome protein degradation pathway compared with HLB-sensitive sweet orange (*Citrus sinensis*) (*7*, *8*). Therefore, we hypothesized that an E3 ligase gene polymorphism between *Citrus* and its distant relatives contributes to the varying resistance to HLB. The citrus genome encodes more than 1400 E3 ligases (data S1). We compared transcriptomes of disease-sensitive *C. sinensis*, disease-tolerant *C. daoxianensis*, and finger lime (*Microcitrus australasica*) to identify potential transcriptionally responsive *S* genes. Several genes encoding plant U-box (PUB)–type E3 showed higher expression in disease-sensitive *C. sinensis* compared with disease-tolerant *C. daoxianensis* and *M. australasica* (Fig. 1B and table S1). Among these, *PUB21* exhibited the highest upregulation during *C*Las infection, although there was no significant transcriptional response in the more HLB-tolerant *M. australasica*.

Positive correlations were observed between *PUB21* expression and disease severity across the Rutaceae family (Fig. 1C) (*6*). Transcript levels of *PUB21* in HLB-susceptible *C. sinensis* were 2.3-fold higher than in *C. daoxianensis* and more than 50-fold higher than in other HLB-tolerant species, such as trifoliate orange (*Poncirus trifoliata*); *M. australasica*; and HLB-resistant distant relatives, such as orange jasmine (*Murraya paniculata*), curry leaf tree (*Bergera koenigii*), Sichuan pepper (*Zanthoxylum bungeanum*), and Rue (*Ruta graveolens*) (Fig. 1C). Down-regulation of *PUB21* correlated with a decrease in *C*Las bacterial titer (Fig. 1D and fig. S1, A and B). We generated stable transgenic citrus plants to test whether *PUB21* alters susceptibility to *C*Las (fig. S1, C to G). Ectopic expression lines (*PUB21* OE) displayed increased HLB symptoms, such as blotchy leaf mottles (Fig. 1, E and F). These lines also exhibited higher *C*Las titers (Fig. 1G), which shows that *PUB21* confers citrus HLB susceptibility.

A PUB21 dominant-negative paralog from *Citrus* relatives confers HLB resistance

A 3500–base pair (bp) genomic region encompassing the promoter and gene-coding sequences of *PUB21* from Rutaceae relatives was analyzed. First, we examined a variable 150-bp sequence insertion of a helitron transposable element in the *PUB21* promoter regions (fig. S2A and data S2). Second, we analyzed four distantly related HLB-resistant Rutaceous species containing an additional PUB21 gene duplication (Fig. 1H and fig. S2B). This PUB21 syntenic paralog contains a point mutation that changes cysteine-39 to alanine in the conserved U-box domain, creating a putative dominant-negative mutant similar to known PUB mutants that disrupt E3 activity (*9*, *10*). This version, named *PUB21DN*, is the predominant *PUB21* paralog expressed in HLB-immune *B. koenigii*, and it is a minor paralog expressed in HLB-resistant *M. paniculate* (fig. S2C). These findings suggest a positive correlation between *PUB21DN* expression and HLB resistance.

Both transient and stable ectopic expression of *PUB21DN* in citrus conferred resistance against HLB infection, as evidenced by phenotypic observations and reduced *C*Las titers (Fig. 1, D to G, and fig. S3A). Ectopic *PUB21DN* expression in transgenic citrus plants rendered them completely immune to HLB when inoculated with *C*Las by psyllid vectors (fig. S3B). A gene-editing vector was used to transform *PUB21* into *PUB21DN* in citrus protoplasts (fig. S3, C and D). This transformation enhanced citrus cell resistance to *C*Las (Fig. 1I). Therefore, PUB21DN enhances citrus resistance to HLB.

MYC2 is a proteolytic substrate of PUB21

PUB21, but not PUB21DN, demonstrated autoubiquitination activity (Fig. 2A). A yeast two-hybrid screen was conducted to identify potential substrate proteins of PUB21. Among the targets, MYC2 is a key transcription factor in plant defense signaling (table S2) (*11*, *12*). Full-length PUB21 and its armadillo repeats bound to MYC2 (Fig. 2, B and C, and fig. S4A). PUB21 interacted with MYC2 in both the cytoplasm and nuclei of citrus cells, consistent with the subcellular localization of PUB21 (fig. S4, A and B). The ubiquitination assays confirmed that PUB21 efficiently polyubiquitinated MYC2, whereas PUB21DN showed reduced ubiquitination of MYC2 (Fig. 2, D and E, and fig. S4C). These findings establish MYC2 as a ubiquitination substrate of PUB21.

PUB21DN was capable of heterodimerizing with PUB21 (fig. S5); thus, we hypothesized that PUB21DN titrates out PUB21 and blocks E3 ligase activity. We found that increasing

[1]State Key Laboratory of Microbial Diversity and Innovative Utilization, Institute of Microbiology, Chinese Academy of Sciences, Beijing, China. [2]CAS Center for Excellence in Biotic Interactions, University of Chinese Academy of Sciences, Beijing, China. [3]BGI Research, Beijing, China. [4]Guangxi Key Laboratory of Agro-environment and Agro-product Safety, College of Agriculture, Guangxi University, Nanning, China. [5]College of Life Science, Gannan Normal University, Ganzhou, China. [6]National Citrus Engineering Research Center, Citrus Research Institute, Southwest University, Chongqing, China. [7]Department of Plant Sciences, University of California, Davis, CA, USA.
*Corresponding author. Email: zhaopz@im.ac.cn (P.Z.); wangxuefeng@cric.cn (X.W.); jianye@im.ac.cn (J.Y.)
†These authors contributed equally to this work.

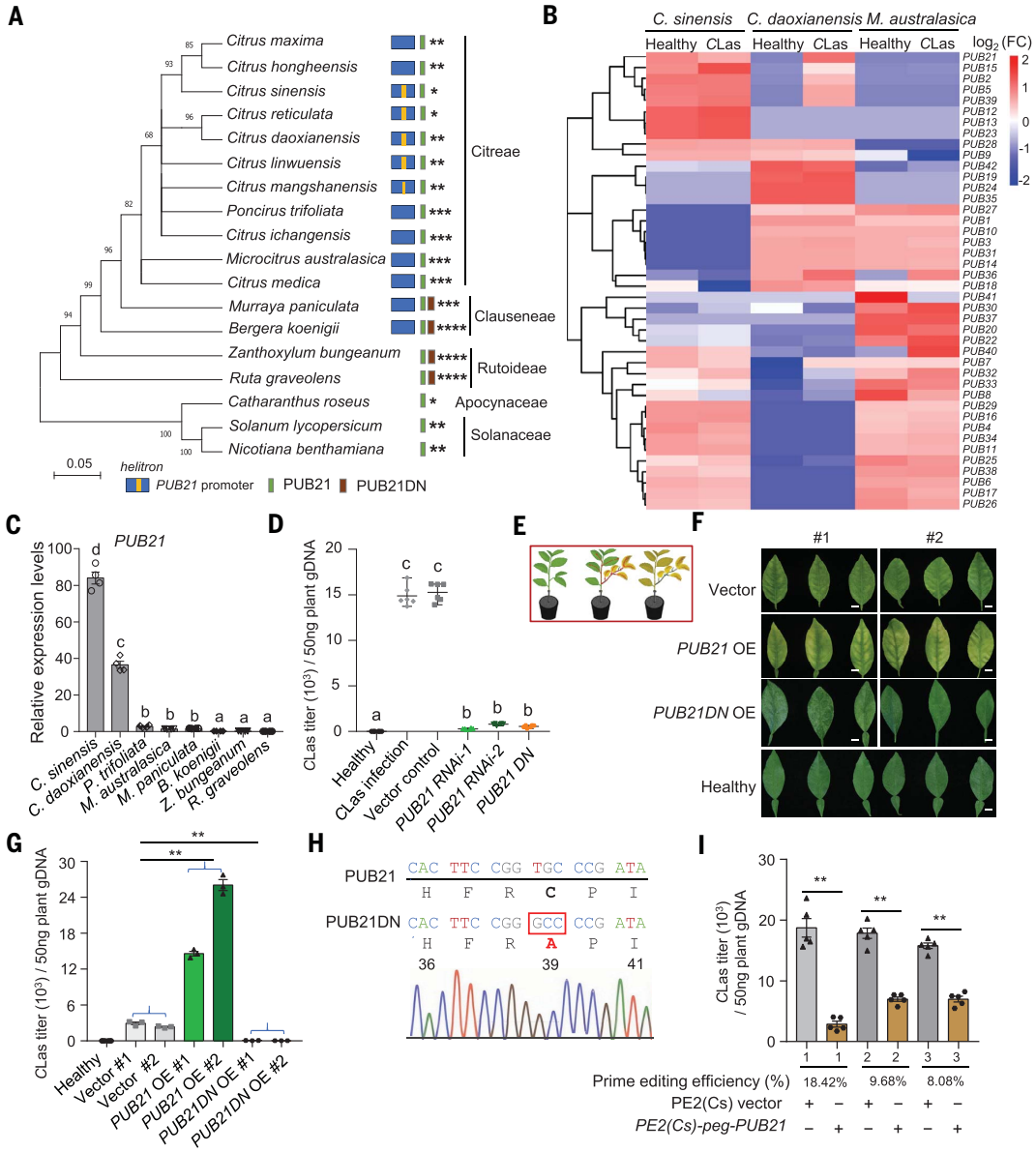


Fig. 1. Regulation of HLB resistance by a U-box E3 ubiquitin ligase PUB21 and its dominant-negative paralog PUB21DN. (**A**) Neighbor-joining phylogenetic tree of Rutaceae species and other species. The numbers of asterisks indicate the HLB resistance level of each species, and orange bars indicate the presence of a helitron transposon in *PUB21* promoters. Green and brown bars indicate whether *PUB21* or *PUB21DN* was present. (**B**) Transcriptomic analysis of *plant U-box genes* (*PUBs*) family in *Citrus* species in response to *C*Las infection. FC, fold change. (**C**) Relative expression levels of *PUB21* gene in Rutaceae species. Data represent means ± SEMs (n = 4 samples). Lowercase letters indicate significant differences among different species (P < 0.05; Duncan's multiple comparisons test). (**D**) The *C*Las amount in transiently *PUB21* RNA interference (RNAi) or *PUB21DN* overexpression of *C*Las-infected *C. sinensis* leaves. Data represent means ± SEMs (n = 6). Lowercase letters indicate significant differences among different lines (P < 0.05; Duncan's multiple comparisons test). gDNA, genomic DNA. (**E**) Schematic diagrams of *C*Las infection by grafting *C*Las-infected scion onto healthy transgenic *C. sinensis* rootstock. (**F**) Leaf symptoms of various transgenic citrus after *C*Las infection for 12 months. Scale bars, 0.5 cm. (**G**) The *C*Las amount in grafting *C*Las-infected transgenic *C. sinensis* overexpressing *PUB21* or *PUB21DN*. Data represent means ± SEMs (n = 3) (**P < 0.01; Student's t test). (**H**) PUB21 and its dominant-negative paralog PUB21DN in distant *Citrus* relatives. The sequence of polymerase chain reaction (PCR) amplicon that contains the C39 locus shows two mutant nucleotides (in red color). (**I**) The *C*Las amount in the empty PE2(Cs) vector and PE2(Cs)-peg-PUB21 transfected HLB-positive citrus protoplasts. Data represent means ± SEMs (n = 5) (**P < 0.01; Student's t test).

PUB21DN levels relative to PUB21 levels inhibited the E3 activity up to near complete inhibition (Fig. 2E). MYC2 ubiquitination concomitantly decreased as the PUB21DN/PUB21 ratio increased. HLB-tolerant relatives, particularly *M. australasica* and *B. koenigii*, exhibited higher MYC2 levels compared with HLB-susceptible *C. sinensis* (Fig. 2F and fig. S6A). To test whether PUB21 posttranslationally controls MYC2 levels, we analyzed MYC2 degradation rates after inhibiting protein synthesis with cycloheximide (Fig. 2, G and H, and fig. S6, B and C). MYC2 in vector control plants had a half-life of ~6 hours, whereas MYC2 in *PUB21* OE lines had a shorter half-life of less than 3 hours. By contrast, MYC2 degradation was significantly slower in HLB-resistant *PUB21DN* OE lines. These results demonstrate

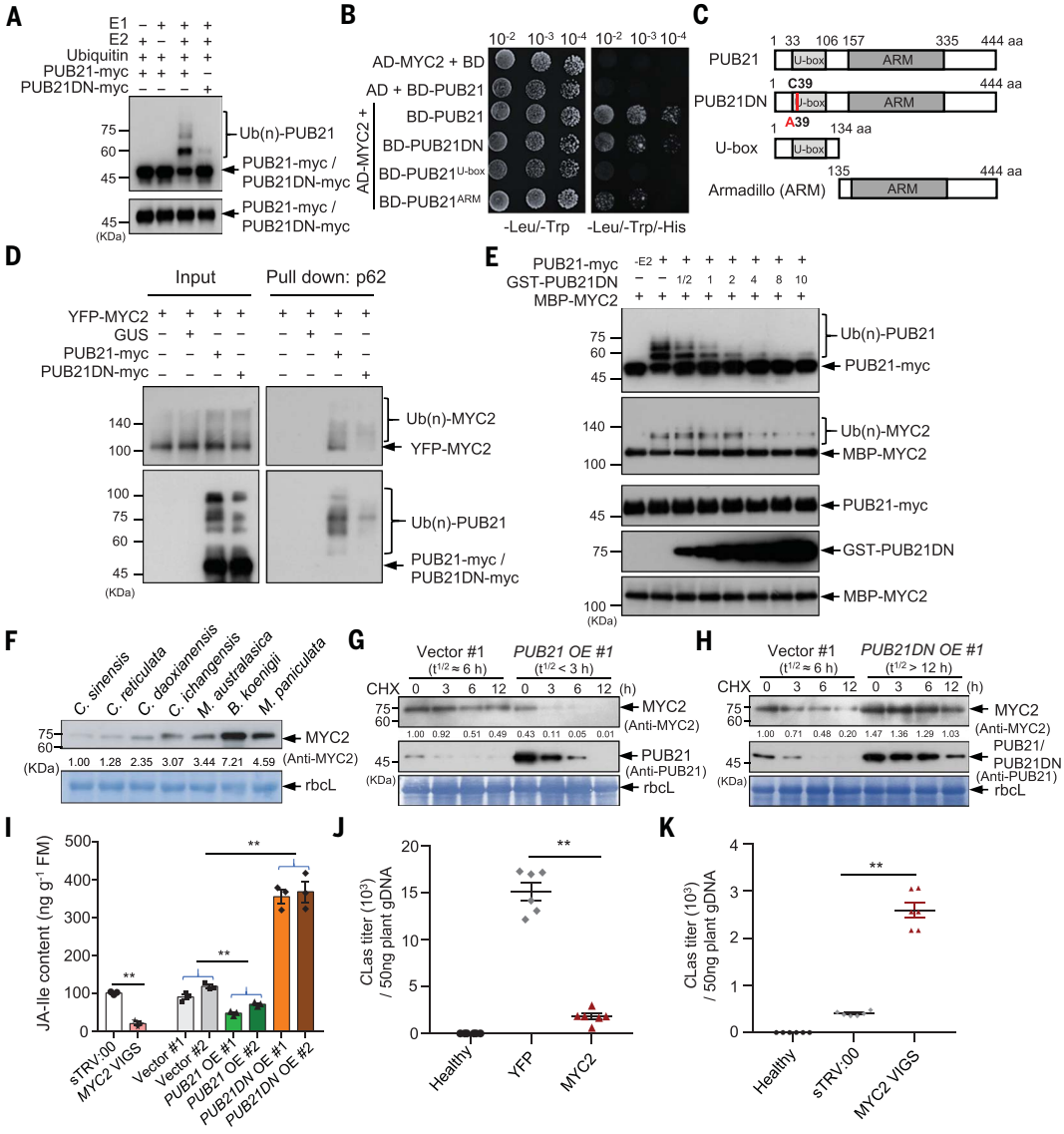


Fig. 2. PUB21 directly targets an HLB-resistant MYC2 for protein degradation. (**A**) PUB21DN with a mutation in the U-box domain abolished its E3 ligase activity. (**B**) Interaction between PUB21 and MYC2 in the yeast two-hybrid system. (**C**) Schematic diagrams of wild-type PUB21 and its deletion derivatives. (**D**) The ubiquitination of MYC2 by PUB21 in planta. (**E**) Excess PUB21DN protein functions as a dominant negative mutant by blocking PUB21 E3 activity. Numbers indicate the relative amounts of proteins in the reaction mixture, where 1 represents 50 ng of PUB21-myc and GST-PUB21DN. (**F**) Endogenous MYC2 levels in Aurantioideae species. (**G** and **H**) Degradation rates of MYC2 in transgenic citrus overexpressed *PUB21* and *PUB21DN*. Transgenic citrus leaves overexpressing *PUB21* (line 1) (G) and *PUB21DN* (line 1) (H) were incubated with 200 μM cycloheximide (CHX) for the indicated time. The normalized value at 0 hours in vector 1 was set to 1.00. (**I**) Levels of endogenous JA-Ile in *MYC2*-silenced citrus and transgenic citrus overexpressing *PUB21* and *PUB21DN* lines. Values are means ± SEMs (n = 3) (**P < 0.01; Student's t test). (**J**) The *C*Las amount in transient *MYC2* overexpression of *C*Las-infected citrus leaves. Data represent means ± SEMs (n = 6) (**P < 0.01; Student's t test). (**K**) The *C*Las amount in *MYC2*-silenced citrus was measured. Values are means ± SEMs (n = 6) (**P < 0.01; Student's t test).

that PUB21DN inhibits PUB21-mediated proteolysis of MYC2.

The PUB21-MYC2 module activates jasmonate-mediated defense

HLB-tolerant species with elevated MYC2 showed higher jasmonate (JA) levels than HLB-susceptible *C. sinensis*, indicating the role of citrus MYC2 in JA signaling (fig. S7A). Confirming this role, *MYC2*-silenced citrus had lower levels of JA and its active form jasmonyl-L-isoleucine (JA-Ile) compared with those in the sTRV vector control citrus (Fig. 2I and fig. S7, B to D). Correspondingly, JA-related biosynthetic and regulatory genes were down-regulated in the *MYC2*-silenced citrus (fig. S7E). *MYC2* silencing also reduced the expression of several genes related to antibacterial metabolite biosynthesis, including isoprenoid synthesis, flavonoid synthesis, and phenylpropanoid metabolism (fig. S7E) (*8*). Furthermore, *MYC2* overexpression decreased *C*Las concentrations, whereas *MYC2* down-regulation increased *C*Las concentration (Fig. 2, J and K). Thus, citrus MYC2 enhances JA-mediated defenses, contributing to resistance against the HLB pathogen.

As expected from the PUB21-MYC2 interaction, *PUB21* OE lines, which decrease MYC2 levels, lead to decreased JA accumulation and downstream gene expression (Fig. 2I and fig. S7, F to H). Expressing *PUB21DN* had the opposite effect and increased JA biosynthesis and downstream gene expression (Fig. 2I and

fig. S7, F to I). Therefore, the PUB21-PUB21DN module regulates MYC2 levels and multiple aspects of JA-mediated resistance to HLB.

Helitron transposition generates a regulatory circuit between PUB21 and MYC2

Comparison of the *PUB21* promoters across Citreae and Clauseneae identified an indel polymorphism caused by the helitron insertion. Helitrons are repetitive single-stranded virus-like DNA transposons, which substantially affect genome variability by capturing and mobilizing host genomic sequences (*13*). There are three forms of *PUB21* promoters: non-helitron, helitron, and an intermediate type (Fig. 3A and data S3). The helitron contains five G-box–like motifs (CANNTG) bound by MYC2 (*11*), creating potential variation in the ability of MYC2 to regulate PUB21. The helitron-containing promoters have 13 MYC2 cis-elements and are present in HLB-sensitive species *C. sinensis* and *Citrus reticulata*. The intermediate types appear to be losing three motifs and have 10 MYC2 cis-elements. The HLB-resistant distant relatives and most HLB-tolerant citrus species belong to the non-helitron type and have only eight MYC2 cis-elements in their *PUB21* promoters.

To assess the effect of helitron and associated MYC2 cis-elements on the *PUB21* expression, a helitron loss-of-function mutant of *C. sinensis PUB21* promoter ($CsPUB21^{\Delta hel}$) was generated. We also generated cis-element swapping mutants by insertion of two MYC2 cis-elements from each of the intermediate type *Citrus*, either 148 bp of *Citrus linwuensis* (cl148) or 87 bp of *C. mangshanensis* (cm87), into $CsPUB21^{\Delta hel}$ to form $CsPUB21^{\Delta hel+cl148}$ and $CsPUB21^{\Delta hel+cm87}$. In transient expression assays, the transactivation activity of MYC2 on the $CsPUB21^{\Delta hel}$ promoter was lower compared with the intact *CsPUB21* promoter (Fig. 3B and fig. S8). Additionally, the lack of helitron insertion in *M. paniculata* and *B. koenigii* resulted in reduced *PUB21* transactivation by MYC2. The intermediate type of insertion had intermediate transcriptional activity reflecting its intermediate number of MYC2 cis-elements. Inserting helitron into three non-helitron types of the *PUB21* promoters—$CiPUB21^{hel}$ promoter of *Citrus ichangensis*, $MpPUB21^{hel}$ promoter of *M. paniculata*, and $BkPUB21^{hel}$ promoter of *B. koenigii*—increased MYC2 transactivation activity, leading to enhanced *PUB21* expression (Fig. 3C and fig. S9). These results demonstrate that MYC2 cis-elements within helitron insertions form a feedback regulatory circuit linking the *PUB21* promoter PUB21-MYC2, promoting *PUB21* expression, and therefore degrading MYC2.

Chromatin immunoprecipitation (ChIP) assays showed that MYC2 directly binds the proximal end of *C. sinensis PUB21* promoter, encompassing regions I, II, and III, with the helitron insertion specifically located in region II (Fig. 3D). MYC2 exhibited binding affinity for the middle region of the *M. paniculata PUB21* promoter and the distal end of *B. koenigii PUB21* promoter (Fig. 3, E and F). These results suggest that MYC2 exhibits different transcriptional activities on these *PUB21* promoters, consistent with higher *PUB21* expression in *C. sinensis* compared with that in *M. paniculata* and *B. koenigii* (Fig. 1C). Electrophoresis mobility shift assays (EMSAs) confirmed the binding of MYC2 to *C. sinensis PUB21* promoter, especially each G-box–like motif within the helitron insertion (Fig. 3, G and H). These results indicate that MYC2 induces *PUB21* expression by direct promoter interaction.

*C*Las effector SDE5 enhances the PUB21-MYC2 interaction

We tested whether *C*Las deploys Sec-delivered effectors (SDEs) to target the PUB21-MYC2 defense node. Out of 86 putative SDE proteins in *C*Las (*14*), we initially focused on 18 secreted proteins that are smaller than 20 kDa in size, lack transmembrane domains, and are specific to the *C*Las pathogen (table S3). Two of these interacted with MYC2, SDE1 (*15*), and SDE5 (Fig. 4A and fig. S10). SDE5 is conserved among 11 *C*Las strains with 100% sequence identity (fig. S11A). Direct tissue blot immunoassay and Western blotting results confirmed that SDE5 is secreted into the phloem sap of *C*Las-infected citrus (fig. S11, B and C).

Both SDE5 and MYC2 interact with the armadillo domain of PUB21 (Fig. 2B; fig. S4A; and fig. S12, A and B), and SDE5 could enhance the PUB21-MYC2 interaction (Fig. 4, B and C, and fig. S12, C to E). SDE5 also increased PUB21-mediated ubiquitination of MYC2 without altering PUB21 activity (Fig. 4D). Coexpression of SDE5 and MYC2 reduced the half-life of MYC2 to less than 3 hours, compared with ~6 hours in controls coexpressing yellow fluorescent protein (YFP) and MYC2 (Fig. 4E). Similarly, MYC2 degradation was faster in *SDE5*-overexpressing transgenic citrus lines compared with vector control lines (Fig. 4F and fig. S13, A and B). Both *C*Las infection and *SDE5* expression increased *PUB21* levels (fig. S13C). Correspondingly, endogenous MYC2 levels in *C*Las-infected citrus were less than 50% of the levels in healthy plants (fig. S13D). Accordingly, *SDE5* expression led to lower JA levels, decreased JA-associated gene expression, and increased sensitivity to *C*Las compared with vector control plants (Fig. 4, G to I, and fig. S13, E to G). Collectively, SDE5 suppresses JA-mediated defenses, thereby promoting HLB infection.

Antiproteolysis peptides target PUB21 E3 ligase activity to suppress *C*Las

The functions of *S* genes extend beyond serving as compatibility factors for pathogens because the side effects caused by their mutation can be minimized through tissue-specific fine-tuning strategies (*4*). An artificial intelligence (AI)–assisted screening pipeline was developed to identify antiproteolysis peptides (APPs) in a human gut microbiome–derived antimicrobial peptide library (Fig. 5A, table S4, and data S4). Owing to their human origin, these peptides exhibit low toxicity and are likely to face fewer regulatory and consumer barriers (*16*). From this library, ~500 potential PUB21-binding peptides were identified (data S5).

Given that the SDE5 effector promotes PUB21-mediated MYC2 degradation, we sought to identify APPs capable of specifically stabilizing MYC2 by inhibiting PUB21 activity in HLB-positive plants while ensuring that these peptides did not alter MYC2 stability in healthy plants to avoid undesired pleiotropic effects. To achieve this, we synthesized 22 small peptides randomly selected from PUB21-binding candidates (table S5) and tested these peptides for their ability to selectively block PUB21-mediated MYC2 degradation in the diseased citrus plants. Three peptides (APP3, APP9, and APP17) effectively inhibited PUB21 activity only in the presence of SDE5 (Fig. 5B and fig. S14A). These APPs were PUB21 specific because they did not affect the E3 activity of an alternative ligase, PUB10, which interacts and ubiquitinates MYC2 in *Arabidopsis* and *Citrus* (fig. S14, B to D) (*9*). The addition of these APPs also reduced MYC2 polyubiquitination levels (Fig. 5B). To rapidly evaluate the effectiveness of these APPs against the HLB pathogen, live leaf injection assays were conducted on *C*Las-infected citrus leaves. The APPs significantly mitigated *C*Las levels, showing more than 10 times the potency of tetracycline (Tet) (Fig. 5C and fig. S14E).

APP3 exhibits bactericidal activity and stabilizes MYC2

To understand how the APPs function, we focused on APP3 for subsequent analysis. The β strand fragment of APP3 (named APP3-14), rather than the α helix fragment, was the functional motif for anti-*C*Las activity (Fig. 5C and fig. S15A). Microscale thermophoresis and pull-down assays verified the direct interaction between APP3 or APP3-14 and PUB21 (Fig. 5D and fig. S15B). APP3 and APP3-14 showed similar binding affinities for PUB21, with moderate binding affinity (APP3, K_d = 16.96 ± 7.17 μM; APP3-14, K_d = 24.18 ± 7.48 μM) (fig. S15B). Competitive binding assays using excess unlabeled APP3 confirmed the high specificity of APP3 for PUB21 (Fig. 5D). SDE5 strengthened the APP3-PUB21 interaction (Fig. 5D), which explains why APP3 suppresses MYC2 ubiquitination only when *C*Las effector SDE5 is present.

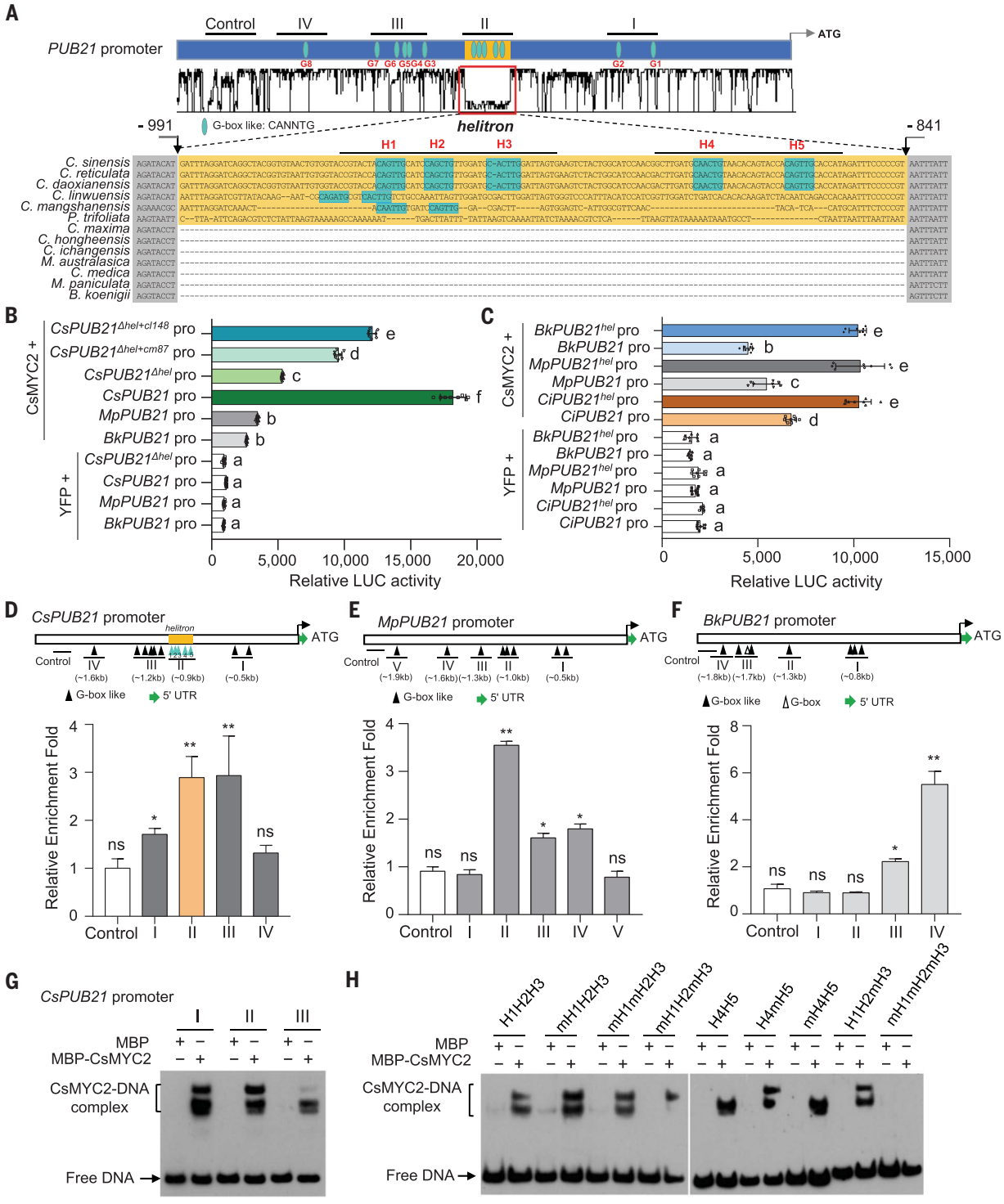


Fig. 3. The helitron insertion causes a regulatory circuit between PUB21 and MYC2. (**A**) Diagram of the helitron insertion in *PUB21* promoter regions of Citreae species. The orange box represents the body of insertions, and the blue boxes represent G-box–like cis-elements. (**B** and **C**) Quantitative luminescence of MYC2 activating the *PUB21* promoter activity in plants. Deletion of the helitron insertion and introduction of intermediate type insertion (B) or introduction of the helitron insertion (C) into non-helitron type of the *PUB21* promoters affected *PUB21* expression. Data represent means ± SEMs (n = 8). Lowercase letters indicate significant differences among different columns (P < 0.05; Duncan's multiple comparisons test). LUC, luciferase. (**D** to **F**) MYC2 directly binds to the *PUB21* promoter. Fold enrichment of MBP-CsMYC2 by coimmunoprecipitation with maltose-binding beads associated with each DNA fragment of *CsPUB21* promoter in ChIP assay (D). MBP, maltose-binding protein; UTR, untranslated region. Fold enrichment of endogenous MYC2 by coimmunoprecipitation with anti-MYC2 resin associated with each DNA fragment of *MpPUB21* promoter (E) or *BkPUB21* promoter (F) in ChIP assay. Data represent means ± SEMs (n = 4) (ns, no significant differences; *P < 0.05; **P < 0.01; Student's t tests). (**G**) EMSA analysis of MYC2 binding to the DNA fragments of *CsPUB21* promoter. (**H**) DNA binding activities of MYC2 protein on each G-box–like motif within the helitron insertion. "H" indicates G-box–like motif, and "mH" indicates the mutant G-box–like motif within the helitron insertion.

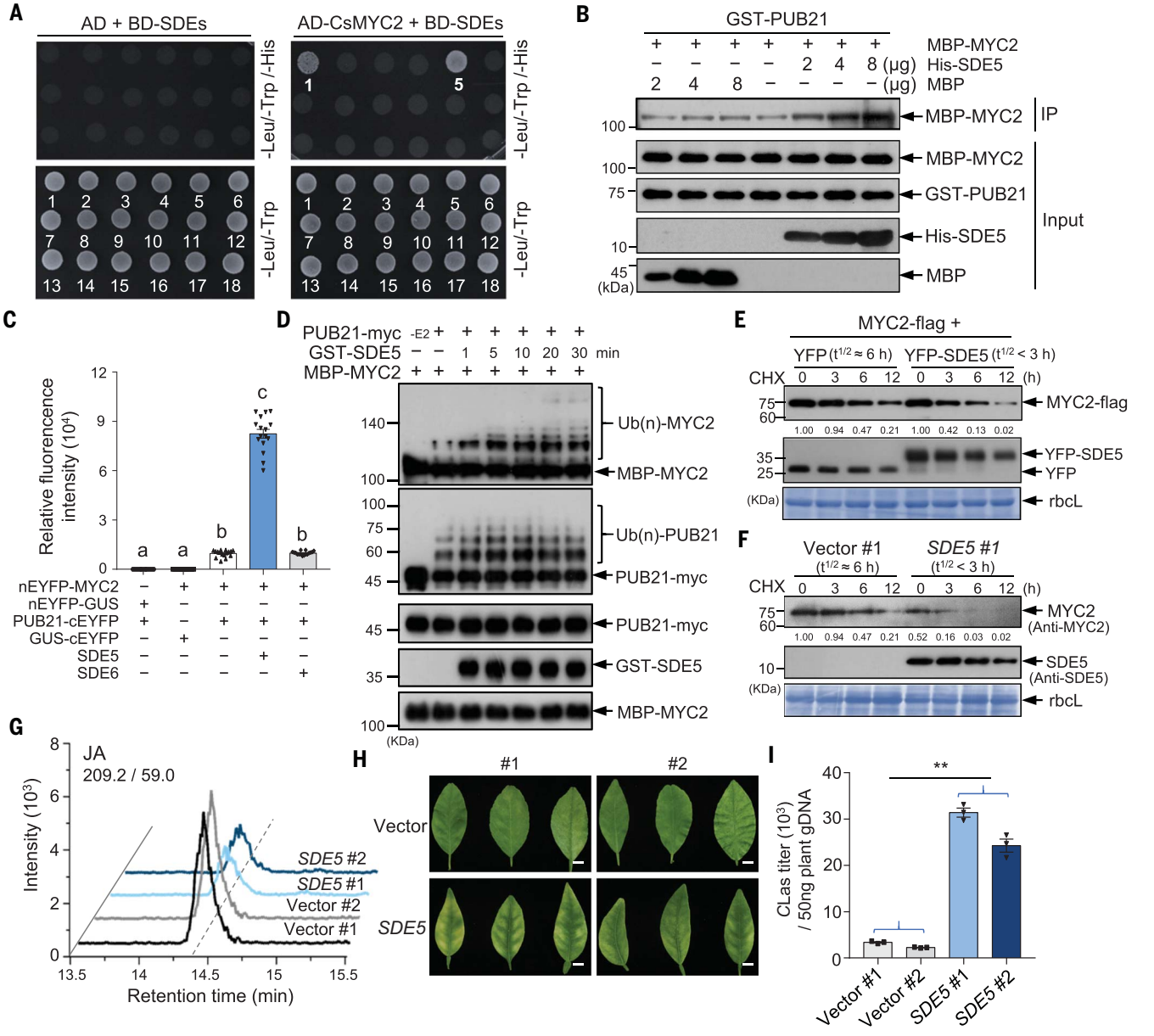


Fig. 4. *C*Las-secreted SDE5 effector enhances PUB21-mediated MYC2 degradation. (**A**) Interaction between SDE effectors and MYC2 in the yeast two-hybrid system. (**B**) Protein competition pull-down assay. The indicated protein amount of His-SDE5 or MBP was mixed with 2 μg of MBP-CsMYC2 and pulled down by 2 μg of GST-PUB21. (**C**) Effects of SDE5 on the interaction of PUB21 with MYC2. Values are means ± SEMs (n = 16). Lowercase letters indicate significant differences among different columns (P < 0.05; Duncan's multiple comparisons test). (**D**) Effects of SDE5 on the ubiquitination of MYC2 protein by PUB21. (**E**) Degradation rate of MYC2 in transiently coexpressing *MYC2-flag* with *YFP* or *YFP-SDE5*. (**F**) Degradation rate of MYC2 in *SDE5*-transgenic *C. sinensis* lines. (**G**) High-performance liquid chromatography–mass spectrometry (HPLC-MS) analysis of endogenous JA in vector control and two *SDE5*-transgenic citrus lines. (**H**) Leaf symptoms of *SDE5*-transgenic citrus lines after *C*Las infection for 12 months. Scale bars, 0.5 cm. (**I**) The *C*Las amount in grafting *C*Las-infected SDE5-transgenic citrus trees for 12 months. Data represent means ± SEMs (n = 3) (**P < 0.01; Student's t test).

To assess whether the anti-*C*Las activity of APP3-14 depends on the PUB21, APP3-14 was applied to citrus plants with silenced or overexpressed *PUB21*. APP3-14 effectively reduced *C*Las levels in all HLB-positive plants (Fig. 5E and fig. S15C). This inhibitory effect was lower in *PUB21*-silenced citrus plants and higher in *PUB21* OE lines compared with the vector control plants, which indicates that the inhibitory effect of APP3-14 on the HLB pathogen is partially dependent on PUB21. APP3 application to *C*Las-infected citrus increased endogenous MYC2 levels, enhancing JA and JA-Ile accumulation and promoting JA-mediated defenses (Fig. 5, F to H, and fig. S15, D to F). MeJA and APP3-14 independently reduced *C*Las levels (fig. S15G). When applied together, MeJA and APP3-14 exhibited an additive inhibitory effect, indicating that APP3-14 suppresses *C*Las partially via the JA pathway by stabilizing MYC2. Transmission electron microscopy (TEM) revealed that APP3 and APP3-14 application to *C*Las caused cytosol leakage and the release of small extracellular vesicles (Fig. 5I). *C*Las cell lysis could be observed frequently within many citrus cells, similar to the antibacterial effects of Tet treatment on the *C*Las pathogen. Taken together, APP3-14 has dual functions—bactericidal activity and anti–PUB21-mediated proteolytic effects.

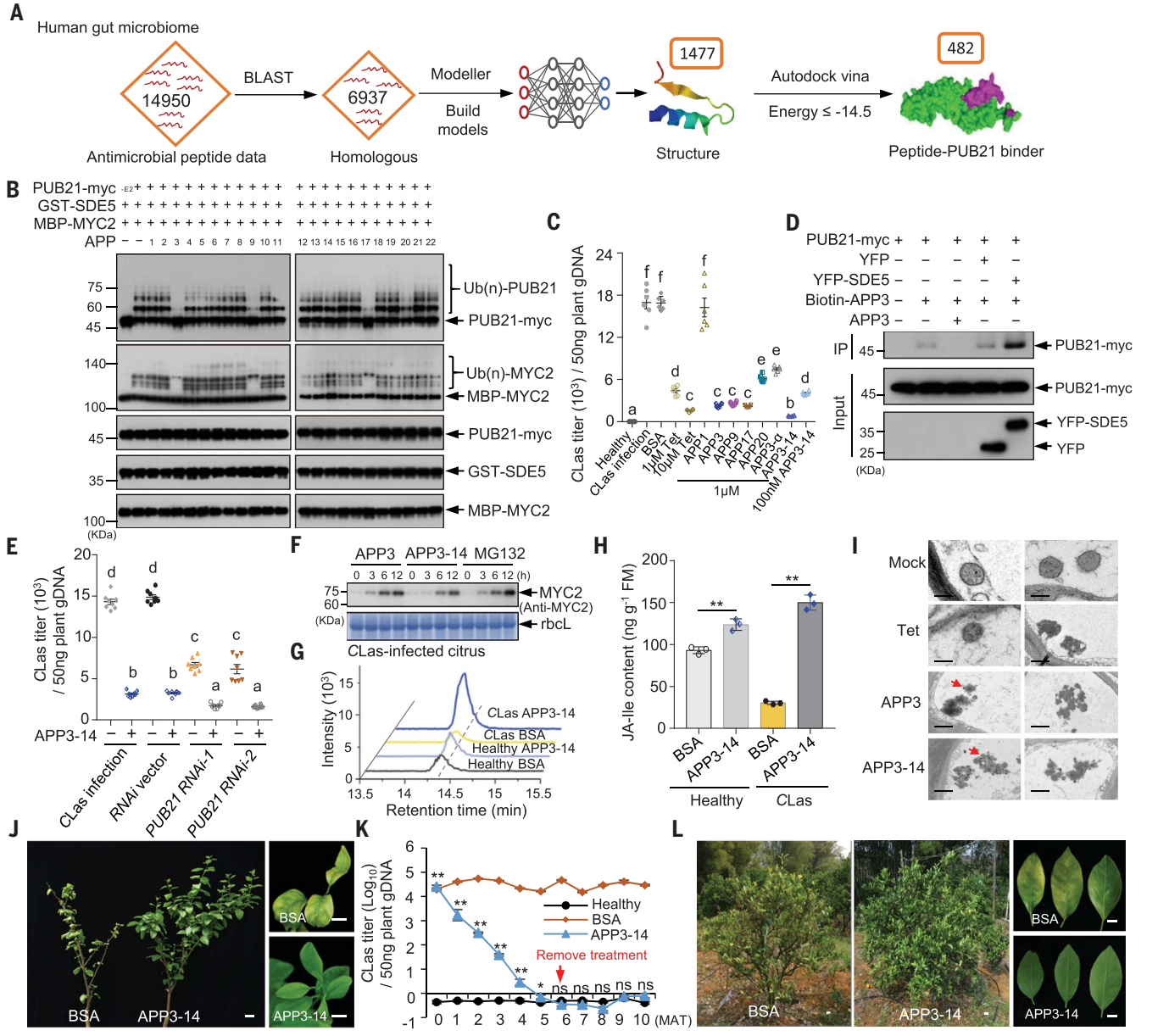


Fig. 5. APPs stabilize MYC2 protein and exhibit bactericidal activity. (**A**) Schematic representation of workflow to identify APPs for inhibiting PUB21 activity. (**B**) Effects of peptides on ubiquitination levels of PUB21 and MYC2 with SDE5 presence. (**C**) The *C*Las amount in *C*Las-infected *C. sinensis* leaves treated by peptides. Values are means ± SEMs (n = 6). Lowercase letters indicate significant differences among different treatments (P < 0.05; Duncan's multiple comparisons test). (**D**) Pull-down assay between PUB21-myc with biotinylated-APP3. (**E**) The *C*Las amount in citrus leaves with or without being treated by APP3-14 peptides applied to *PUB21*-silenced citrus. Values are means ± SEMs (n = 8). Lowercase letters indicate significant differences among different treatments (P < 0.05; Duncan's multiple comparisons test). (**F**) Effects of APP3 and APP3-14 peptides on MYC2 accumulation in *C*Las-infected citrus. (**G**) HPLC-MS analysis of endogenous JA in APP3-14–treated citrus. (**H**) Levels of endogenous JA-Ile in APP3-14–treated citrus. Values are means ± SEMs (n = 3) (**P < 0.01; Student's t test). (**I**) TEM images of *C*Las cells treated with peptides. Red arrows indicate cytosol leakage or vesicle release. Scale bars, 350 nm. (**J**) Symptoms of symptomatic leaves/total *C. sinensis* trees after APP3-14 treatment for 6 months in the greenhouse. Scale bars, 2 cm. (**K**) The *C*Las amount in *C*Las-infected citrus trees treated by APP3-14. MAT, month after treatment. Values are means ± SEMs (n = 6) (ns, no significant differences; *P < 0.05; **P < 0.01; Student's t tests between healthy citrus and APP3-14–treated *C*Las-infected citrus). (**L**) Symptoms of symptomatic leaves/total *C. sinensis* trees after APP3-14 treatment for 4 months in field trials. Scale bars, 2 cm.

Protein stabilization peptides inhibit *C*Las in HLB-positive citrus trees

To determine APP3-14 efficiency on whole HLB-positive citrus trees in the greenhouse, trunk injection was used to deliver APP3-14 into *C. sinensis* trees. Monthly monitoring revealed that APP3-14 treatment significantly reduced disease symptoms, resulting in the substantial growth and development of symptomless new flushes, in contrast to what was observed in BSA-treated trees, which continued to exhibit symptomatic flushes (Fig. 5J and fig. S16A). Bacterial titer in the APP3-14–treated trees decreased from the first month after treatment. These trees had fully recovered to a healthy

status at 6 months after treatment with no detectable *C*Las (Fig. 5K). No *C*Las rebound was observed in the months after the removal of APP3-14 treatment, further supporting the curative potential of APP3-14.

In a small-scale field trial, we applied APP3-14 to an HLB-positive *C. sinensis* plantation in Nanning, Guangxi Province. The results demonstrated that a 4-month APP3-14 treatment could control HLB infection. The treated trees showed reduced disease incidence, including diminished blotchy mottling and yellowing of leaves, alongside the growth of symptomless new flushes starting from 2 months after treatment, with improved outcomes by 4 months after treatment (Fig. 5L and fig. S16, B and C). Additionally, bacterial titers significantly decreased after APP3-14 treatment compared with those in BSA controls (fig. S16D). In summary, APP3-14 represents a viable treatment for HLB by reducing bacterial levels, alleviating disease symptoms, and promoting the recovery of healthy growth.

Discussion

The tribe Citreae comprises the *Citrus* genus and its related genera *Poncirus* and *Fortunella*, all of which are vulnerable to HLB. This study uncovered two *PUB21*-related evolutionary events associated with HLB resistance in distant *Citrus* relatives (fig. S17). First, the evolution of PUB21 dominant-negative (PUB21DN) paralogs within the Rutaceae family likely occurred ~25 million years ago (*7*, *17*). This was when the extensive radiation of angiosperms took place, coinciding with the divergence of *Zanthoxylum*, *Murraya*, *Clausena*, *Glycosmis*, *Luvunga*, and *Aegle* from *Citrus* and predating the collision of the Indian and Asian plates. Secondly, the insertion of helitron into the *PUB21* promoter likely occurred soon after the separation of *Citrus* from sister species *C. ichangensis*, *P. trifoliata*, and *C. mangshanensis* (*7*, *13*). Collectively, the absence of PUB21DN and the creation of an enhanced PUB21-MYC2 regulatory feedback loop created the conditions for susceptibility to HLB.

Targeted protein degradation is a feasible pharmacological strategy to develop antimicrobial drugs (*18*). Our APPs stabilize MYC2 by inhibiting proteolysis, showing antimicrobial potential (Fig. 5). Anti-*C*Las activity of APP3-14 is a result of its β strand motif, which can self-assemble into stable, membrane-disrupting structures, such as β barrel (*19*). However, the mechanisms of these APPs remain to be further elucidated. Despite their promise, challenges, such as peptide delivery, stability, and field efficacy, limit their current agricultural application (*20*). Meanwhile, there are more than 300 bacterial diseases caused by uncultivable or difficult-to-culture pathogens, including *Liberibacter*, *Phytoplasma*, *Spiroplasma*, and *Xylella* (*21*). The strategies developed in this work may help tackle these diseases by targeting either pathogen effector or host immune protein stabilization.

REFERENCES AND NOTES

1. J. H. Graham, R. B. Bassanezi, W. O. Dawson, R. Dantzler, *Annu. Rev. Phytopathol.* **62**, 243–262 (2024).
2. B. Hu *et al.*, *PLOS Pathog.* **17**, e1010071 (2021).
3. C. Y. Huang *et al.*, *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2019628118 (2021).
4. C. C. N. van Schie, F. L. W. Takken, *Annu. Rev. Phytopathol.* **52**, 551–581 (2014).
5. S. Li *et al.*, *Nature* **602**, 455–460 (2022).
6. C. Ramadugu *et al.*, *Plant Dis.* **100**, 1858–1869 (2016).
7. Y. Huang *et al.*, *Nat. Genet.* **55**, 1964–1975 (2023).
8. H. D. Wu, Y. Hu, S. M. Fu, C. Y. Zhou, X. F. Wang, *Physiol. Mol. Plant Pathol.* **109**, 101457 (2020).
9. C. Jung *et al.*, *Plant Cell* **27**, 2016–2031 (2015).
10. R. González-Lamothe *et al.*, *Plant Cell* **18**, 1067–1083 (2006).
11. R. Li *et al.*, *Plant Cell* **26**, 4991–5008 (2014).
12. X. Wu *et al.*, *PLOS Pathog.* **15**, e1007897 (2019).
13. D. Barro-Trastoy, C. Köhler, *Trends Genet.* **40**, 437–448 (2024).
14. S. Prasad, J. Xu, Y. Zhang, N. Wang, *Front. Microbiol.* **7**, 1989 (2016).
15. K. Clark *et al.*, *Nat. Commun.* **9**, 1718 (2018).
16. Y. Ma *et al.*, *Nat. Biotechnol.* **40**, 921–931 (2022).
17. M. Wang, S. Tong, T. Ma, Z. Xi, J. Liu, *Mol. Ecol. Resour.* **21**, 2533–2545 (2021).
18. J. M. Tsai, R. P. Nowak, B. L. Ebert, E. S. Fischer, *Nat. Rev. Mol. Cell Biol.* **25**, 740–757 (2024).
19. A. A. Vorobieva *et al.*, *Science* **371**, eabc8182 (2021).
20. H. Ilyas, A. Datta, A. Bhunia, *Curr. Med. Chem.* **24**, 1350–1364 (2017).
21. W. Huang *et al.*, *Mol. Plant* **13**, 1379–1393 (2020).

ACKNOWLEDGMENTS

We thank Y. Wu [Institute of Microbiology, Chinese Academy of Sciences (IMCAS)] for microscale thermophoresis (MST) experiments and JA analysis and J. Liang (IMCAS) for technical support with the TEM. We thank D. Lv (Shanghai Jiaotong University), J. Liu (China Agricultural University), and N. Xu (China Agricultural University) for support with ubiquitination assays. We thank Z. Liang (Shanxi University) for support with PE2(V2) vector. **Funding:** This study was supported by grants from the National Natural Science Foundation of China (32125032), the China National Key Research and Development Program (2021YFD1400800, 2023YFC2606800, and 2024YFC3406000), the National Natural Science Foundation of China (32090013 and 32072393), the CAS Projects for Young Scientists in Basic Research (YSBR-080), and the Strategic Priority Research Program of the Chinese Academy of Sciences (XDA0450000 and XDB0810000). **Author contributions:** Conceptualization: J.Y., P.Z.; Data curation: J.Y., P.Z., Y.S., H.-X.S., Q.X., K.G., X.C.; Formal analysis: J.Y., P.Z., Y.S., J.Z., C.Y., X.Z., H.Y., C.Zhu; Funding acquisition: J.Y., P.Z., Y.S., C.Zho., X.W.; Investigation: J.Y., P.Z., Y.S., C.Y., X.C., H.X., X.Z., H.Y., J.W., X.W., Q.Liu, Q.Li, Q.X., C.Zhu, K.G., X.C.; Methodology: P.Z., Y.S., C.Y., L.Y., H.Y., H.-X.S.; Project administration: J.Y., P.Z., X.W.; Resources: J.Y., P.Z., X.Z., L.Y., C.Zho., X.W.; Software: H.X., H.-X.S.; Supervision: J.Y., P.Z., X.W.; Validation: P.Z., Y.S., X.Z., H.Y., K.G., J.W., C.Zho.; Visualization: J.Y., P.Z., H.Y., Y.S., J.W., H.Y., X.C., K.G.; Writing – original draft: J.Y., P.Z., H.Y., D.J.K., R.F.; Writing – review & editing: J.Y., P.Z., H.Y., Y.S., H.-X.S., D.J.K., R.F., C.Zho., X.W. **Competing interests:** Patents relating to this work and their usage have been filed by the Institute of Microbiology, Chinese Academy of Sciences (China patents: 202110376305.2, 202310033293.2, 202310241068.8, 202310086555.1, 202510258021.1, and 202510122567.4; WIPO patents: PCT/CN2024/076842, PCT/CN2024/080958, and PCT/CN2025/084478). The authors declare that they have no other competing interests. **Data and materials availability:** All data are available in the main text or the supplementary materials. The RNA sequencing data for transcriptome analysis can be accessed at the National Center for Biotechnology Information (NCBI) under accession nos. PRJNA579742 and PRJNA755969.

License information: Copyright © 2025 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adq7203
Materials and Methods
Figs. S1 to S17
Tables S1 to S6
References (*22*–*40*)
MDAR Reproducibility Checklist
Data S1 to S5

Submitted 28 May 2024; resubmitted 19 December 2024
Accepted 11 February 2025
10.1126/science.adq7203